

Double-click (or enter) to edit

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```
from google.colab import files
files.upload()
```

```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

from sklearn.model_selection import train_test_split
from sklearn.linear_model import LinearRegression
from sklearn.preprocessing import LabelEncoder
from sklearn.metrics import mean_absolute_error, mean_squared_error, r2_score
```

```
data = pd.read_csv("Bangalore_Restaurants.csv")
data.head()
```

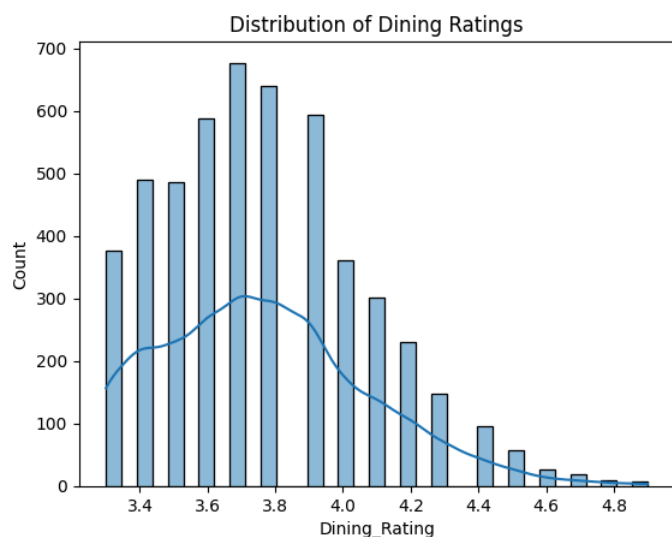
	Restaurant_Name	Category	Pricing_for_2	Locality	Dining_Rating	Dining_Review_Count	Delivery_Rating	Delivery_Rating_Count
0	Burma Burma	Asian, Burmese, Bubble Tea, Salad, Tea, Desser...	1500	Indiranagar, Bangalore	4.9	2790.0	4.5	838.0
1	Windmills Craftworks	Continental, Fast Food, Kebab, Beverages, Ital...	2500	Windmills Craftworks, Bangalore	4.9	6543.0	4.2	524.0
2	CTR Shri Sagar	South Indian	150	Malleshwaram, Bangalore	4.9	4837.0	4.3	22100.0
3	Brahmin's Coffee Bar	South Indian	100	Basavanagudi, Bangalore	4.9	2975.0	4.4	372.0
4	Milano Ice Cream	Desserts, Ice Cream, Beverages	400	Indiranagar, Bangalore	4.9	2575.0	4.4	1180.0

```
data.shape
data.info()
data.describe()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 5109 entries, 0 to 5108
Data columns (total 13 columns):
#   Column                Non-Null Count  Dtype
---  -
0   Restaurant_Name        5109 non-null  object
1   Category               5109 non-null  object
2   Pricing_for_2          5109 non-null  int64
3   Locality               5109 non-null  object
4   Dining_Rating          5101 non-null  float64
5   Dining_Review_Count    5101 non-null  float64
6   Delivery_Rating        3697 non-null  float64
7   Delivery_Rating_Count  5101 non-null  float64
8   Website                5109 non-null  object
9   Address                5109 non-null  object
10  Phone_No               5109 non-null  object
11  Latitude               5109 non-null  float64
12  Longitude              5109 non-null  float64
dtypes: float64(6), int64(1), object(6)
memory usage: 519.0+ KB
```

	Pricing_for_2	Dining_Rating	Dining_Review_Count	Delivery_Rating	Delivery_Rating_Count	Latitude	Longitude
count	5109.000000	5101.000000	5101.000000	3697.000000	5101.000000	5109.000000	5109.000000
mean	568.790370	3.765340	276.544011	3.835110	1655.631249	12.971842	77.584676
std	508.652835	0.304306	873.119616	0.292891	4491.941931	0.498430	1.466164
min	100.000000	3.300000	0.000000	2.500000	0.000000	0.033482	0.000000
25%	300.000000	3.500000	13.000000	3.700000	12.000000	12.918861	77.578063
50%	400.000000	3.700000	46.000000	3.900000	147.000000	12.964121	77.614647
75%	600.000000	3.900000	194.000000	4.000000	1073.000000	13.000115	77.651366
max	6000.000000	4.900000	25500.000000	4.800000	75700.000000	37.249009	77.815213

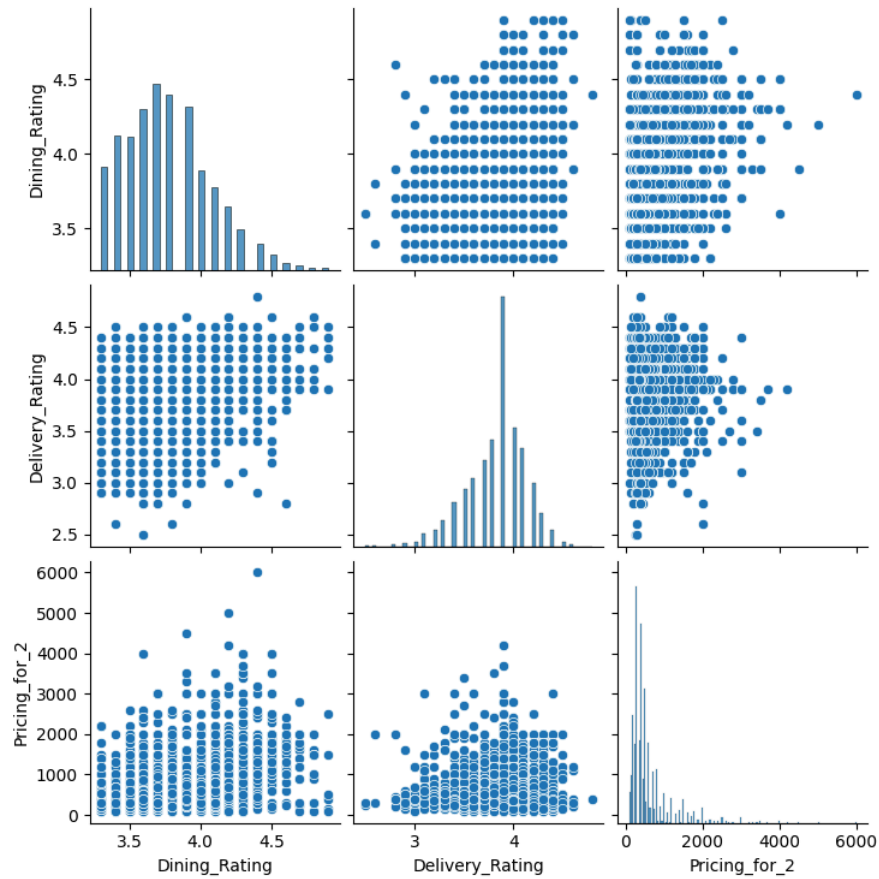
```
sns.histplot(data['Dining_Rating'], kde=True)
plt.title("Distribution of Dining Ratings")
plt.show()
```



```
sns.lmplot(
    x='Pricing_for_2',
    y='Dining_Rating',
    data=data,
    height=6
)
plt.title("Price vs Dining Rating")
plt.show()
```



```
sns.pairplot(data[['Dining_Rating', 'Delivery_Rating', 'Pricing_for_2']])
plt.show()
```



```
corr = data[['Dining_Rating', 'Delivery_Rating', 'Pricing_for_2']].corr()
sns.heatmap(corr, annot=True, cmap='Wistia')
plt.title("Correlation Heatmap")
plt.show()
```

